

Tanvir Singh

AI / ML Engineer | Agentic RAG, Data Pipelines, Python

+61451616111 · tanvir.s2k1@gmail.com · linkedin.com/tanvirsingh01 · Remote – Bengaluru, Karnataka

Summary

Tanvir Singh is an AI and ML practitioner with experience building agentic RAG systems, schema-aware retrieval tools, and data pipelines for real-world query workflows.

Experience

Freelance AI Expert — CosX

Jan. 2026 – Present · Remote – Bengaluru, Karnataka

- Architected hierarchical multi-agent system using LangGraph to orchestrate specialised workers.
- Developed Agentic RAG pipeline with multi-hop reasoning and self-correction.
- Engineered schema-aware retrieval toolsets translating ambiguous NL into optimised SQL.

Research Assistant – Dept. of Computer Engineering & Technology — Guru Nanak Dev University

Apr. 2023 – Jan. 2024 · Amritsar, Punjab

- Benchmarked Topic Modeling architectures (LDA, BERTopic, NMF) for granular text insight.
- Engineered NLTK/SpaCy preprocessing reducing noise 35%.
- Co-authored a formal research paper; managed LaTeX typesetting.

Education

Queensland University of Technology

Master of Information Technology (Data Science) · GPA 5.90/7.00 · Brisbane · Nov. 2026 (Expected)

Guru Nanak Dev University

B.Tech, Computer Science & Engineering · GPA 8.69/10.00 · Amritsar · Oct. 2020 – June 2024

Skills

Programming Languages: Python, C++, Java, R

AI & ML Engineering: LLMs, RAG, LangChain, LangGraph, LangSmith, TensorFlow, PyTorch

Tools & MLOps: pytest, Pydantic, CI/CD, GitHub, GitLFS, AWS, LaTeX

Projects

MediBridge

LangGraph, LangChain, ChromaDB, SQLite (FTS5), OpenAI, Pydantic

- Built agentic RAG chatbot for Australian OSHC queries with 6 LangChain tools over dual-store knowledge base.
- Designed hybrid retrieval fusing dense vectors and SQLite FTS5 via Reciprocal Rank Fusion (60/40).

Activities & Leadership

Project Director — The Emerging Coders Hub (TECH), QUT

Apr. 2024 – Present

- Directed a student tech club delivering technical workshops for 200+ developers.
- Built strategic relationships with industry speakers and corporate sponsors.

Tech Head — Computer Engineering Student Society (CESS), GNDU

Feb. 2022 – Nov. 2024

- Oversaw technical roadmap; managed digital infrastructure for the engineering student body.
- Mentored junior committee members on best practices and project management.